



RIBA PART 3
EXAMINATION IN PROFESSIONAL PRACTICE

**February 2015
Practice Problems**

Day 1

Questions 1-5

GUIDANCE NOTES

In order to be registered as an architect under the Architects Act 1996 and to be eligible for chartered membership of the Royal Institute of British Architects, candidates have to be able to demonstrate their ability to deal competently with situations that arise in the ordinary course of general architectural practice. The practice problems of the RIBA Part 3 Examination in Professional Practice are devised in order to provide candidates with the opportunity to demonstrate their professional competence, working in realistic conditions over two days in their office.

Candidates are to imagine themselves as an employee of the architectural practice described in the scenario, taking the role of the Candidate, responding to each of the problems set out for each of the two days, using the information given in the scenario and in the particular problems. All of the questions carry equal weight and each of them must be attempted. The questions are issued at 0900 to the candidates by the nominated Examination Supervisor each day.

Some of the questions may well require more investigation than others and the responses to the problems may well vary in length and complexity depending on the nature and circumstances involved. It is suggested that the Candidate first read through all of the questions before allocating an appropriate amount of time to each of the problems and allowing sufficient time to read through their responses before depositing them with their Examination Supervisor by 1730 of each day.

Normally the total amount of material submitted in response to the problems is not expected to exceed 5,000 words. Where reference is made to published material the source and the page number must be cited; the copying of published material is not required and should not be necessary information gained from websites must also be cited. Clear succinct responses are preferred to over long verbose discursive answers. Candidates are advised to ensure that their answers are in the form requested, for example where a question asks for a letter it is essential that the response is in the form of a conventional business letter.

Responses to the problems must be typed or printed on A4 size sheets of white paper.

The information given in the scenario and in each of the questions is designed to be helpful and relevant; it is strongly recommended that it should be read carefully before making responses to the problems. Any assumptions made in the course of responding to a problem must be indicated and candidates must be prepared to discuss them during the professional interview.

Candidates may use any reference material, standard documents, the office library and the normal office aids including computers, word processors, photo-copiers, stationery, standard documents within the office, subject to the agreement of your employer. The practice problems and the responses to them must not be discussed with others until after their submission which must include a declaration notice signed by the candidate and the candidate's examination supervisor certifying that it is the candidate's own unaided work.

Where there is more than one candidate taking the examination within the same office, reference material may be shared but candidates may not discuss the problems or their responses to them until after their submission.

Not later than 1730 at the end of each of the two days (unless candidates have been granted additional 2 hours), candidates are required to provide their Examination Supervisor with photocopies or hard copy computer printouts of their draft responses in sealed envelopes which are then signed and retained until their return on the third day for inclusion in the documentary submissions. The copies of the responses are devised as a means of protecting the integrity of the examination; they may be used by the examiners in comparing them with the final versions in the documentary submission.

The third day is solely for the copy typing, proof reading and typographical correction, collation and spiral binding of two copies of the documentary submission which have to be dispatched together with the signed envelopes containing the Day One and Day Two draft responses, the Examination Supervisors Declaration certifying that the whole submission is the candidates own unaided work and a CD of your submission.

The parcel containing these documents must be securely packed and carefully addressed to Alison Mackinder, Director RIBA Part 3, RIBA NW Region, Unit 101, The Tea Factory, 82 Wood Street, LIVERPOOL, L1 4DQ.

Parcels must be dispatched by 1730 on the third day using the Post Office Guaranteed Delivery System or its equivalent and a timed receipt obtained for submission to the examiners at the professional interview. Alternatively submissions may be delivered to the RIBA NW Regional Office prior to 1730.

The cover of the documentary submission must show the candidate's full name, the title of the practice in which the responses to the problems were written and the name of the Examination Supervisor.

Candidates are advised to prepare a copy of their submission for reference in preparation of the professional interview and if need be in the course of the interview.

SCENARIO

The name NDG Architects Ltd is listed in the current RIBA Directory of UK Chartered Practices.

Established in 1998, the practice has a wide range of design and development experience across a number of private and public sector areas, including commercial, retail, industrial, education and health care.

Exclusively based in the North West of England, NDG Architects' office is located in a converted mill building in Central Manchester.

The practice has noticed a sustained upturn in its workload, particularly educational projects, over the past year which has allowed it to forecast the next 6 months with renewed confidence.

As a candidate for the February 2015 examination you are required to assume that you are a salaried employee of NDG Architects and have been engaged at this practice for a period of 20 months whilst preparing to sit the Part 3 examination.

Companies House records show that NDG Architects Ltd has two directors and two associate directors.

James Noble has suffered health problems recently and has been away from the office for a period. He remains as a consultant working part time and is an important link to some longstanding Clients.

Belinda Carter architect and director. Her supplementary role within the company is to maintain the marketing operations, which prior to 2011 had generated significant new multi million pound commissions for blue chip clients. Belinda has now assumed the role of MD but has made it clear that over the next year she feels that significant restructuring will have to take place within the practice.

Michael Hart associate director, RIBA specialist conservation architect and listed on the Diocese schedule of approved architects. His reputation in this field has previously resulted in the practice being commissioned to carry out larger conservation projects attracting lottery funding. Michael is also listed as a Client Design Advisor on the RIBA's CDA register.

Sue Allen, architect and associate director is an accredited sustainability expert and has recently completed a MSc course in Advanced Sustainability at the University of Madchester. Sue also acts as the company's quality manager, responsible for the ongoing development and maintenance of the quality assurance system and manual.

Certain significant projects currently and previously handled by the office:

Conversion of farm buildings at Tally Ho Manor

The practice has been involved with this project for a number of years, master-planning the site and completing refurbishment works at the Manor house. Following recent successful projects the Client, Lord Avafewbob, is looking to maximize the potential of the remaining redundant farm buildings on the site.

Madchester Academy

The practice has undertaken work for many years for Madchester Academy, the most recent project being a £2m community performing arts center which has been given the go ahead by the board of Governors.

Madchester University

The University has a number of sports related projects going forward at the moment. All their consultant appointments are as a result of competitive tender with a 60/40% cost / quality assessment. Fortunately NDG has a good track record of delivering successful projects.

Luxury Houses and apartments at Tracy Fields

The practice has recently been working for a rather aggressive developer client, Mr Hood, who is keen to maximize the financial potential of every site. Some of the developments are in very sensitive locations. The practice is keen to work with this new Client but is currently feeling its way.

ABC Properties

This landlord has a large number of small properties locally and NDG provide a service as required dealing with minor refurbishment, repair and remodeling projects.

Remodelling of Victorian house for Lord and Lady Alltite

The project has been on the books for rather too long with this very demanding Client making various changes in order to meet their varying needs. It has finally gone out to tender and the office is looking forward to getting it off the books.

Luxury Apartments in Madchester

In October 2014 the practice was given the go ahead to proceed with this project which had been on hold since the completion of stage 3. Work started on site at the beginning of the year under a JCT Standard Building Contract with Quantities 2011.

Staff and salaries/director remuneration

James T. Noble	Director	£ 52,000.00
Belinda C. Carter	Director	£ 52,000.00
Michael Hart	Associate Director	£ 38,000.00
Sue Allen	Associate Director	£ 38,000.00
Dareem Patel	Architect	£ 35,000.00
Sven Berg	Architect	£ 35,000.00
Tomas Hjeltnes	Architect	£ 31,000.00
Renata Wang	Architect	£ 31,000.00
George Gee	Architectural Technologist	£ 27,000.00
Candidate	Architectural Assistant	£ 24,000.00
Olga Poulicek	Architectural Assistant	£ 24,000.00
Mark Jones	Architectural Technician	£ 21,000.00
Liv Grosfjelt	Year out student	£ 15,000.00
Sara Smitt	Year out student	£ 15,000.00
Sarah Jane Key	Administrative assistant	
Sophie Bright	Receptionist/ Clerical assistant	
Jane Dudley	Clerical assistant	
Anne Holding	Accounts assistant	

NB All of the characters, locations and projects of the scenario and the practice problems are entirely fictitious and any similarity or resemblance to persons, places, projects or practices is totally coincidental.

End

QUESTION 1

Page 1 of 2

Email@

Date: 25th February 2015

To: Candidate@ndgarchitects.com

From: Belinda@ndgarchitects.com

Re: Tracy Fields Development – DG Hood Developments

As you know we have just completed RIBA stage 2 design work on the housing scheme for our developer client, Mr Hood, whose business is now expanding rapidly. He expects us to 'fast track' the project to keep pace with his growing business.

The Tracy Fields Project is on a very sensitive site in a conservation area with a G1 Listed house on the edge; it also has a copse of historic trees covered by a blanket TPO. There are well established, active, conservation societies in the area, who keep an eye on unwelcome change and they have already got the local media involved.

We have drawn the maximum possible number of units as requested by Mr Hood and are now in the process of trying to convince the Planning and Conservation officers as part of our pre app negotiations, of the merits of the scheme (Mr Hood is suggesting we do not show all the trees on our drawings at this stage).

As you know, you and I are not actually part of the team within the office dealing with this job, but I do feel rather uneasy about it since there appears to be double the density that might appear sensible, and really, the quality of design that has been possible within the tight deadline is rather mediocre. I am concerned that if the project were to be approved, it might affect the standing of the office in the local community (or worse).

The job will be reviewed at our monthly directors meeting and I want to get my thoughts straight. As you know Mr Hood is an important Client and accounts for a good proportion of our fee income, nevertheless, I am concerned we risk behaving unprofessionally.

QUESTION 1
Page 2 of 2

In order to assist me, could you please draft a short list of bullet points on key issues which arise from what he is expecting us to do, together with the adverse consequences for the scheme and the practice.

How do you see our broader responsibilities to the environment and society generally?

Are there any issues covered by the RIBA and ARB codes?

Finally, could you draft a carefully worded letter to Mr Hood on my behalf; we don't want to lose him as a client, but we cannot proceed at any price professionally. You do not need to refer in any detail to planning/conservation legislation that will apply; I will be dealing with that in a separate letter.

Many thanks

Belinda Carter
Director

ndg architects ltd

QUESTION 2
Page 1 of 1

Email@

Date: 25th February 2015

To: Candidate@ndgarchitects.com

From: Michael@ndgarchitects.com

Re: Victorian House remodelling for Lord & Lady Alltite

I don't quite know what has happened here but the tenders for Lord Alltite's house interiors have been returned and are 50% in excess of the budget he gave us initially.

He's going to be asking why !

I would like to respond firmly but VERY diplomatically and also positively.

It seems to me that there are two probable causes:

- The changes in content and quality that Lady Alltite has inevitably asked for during the design process
- The market conditions prevailing now as opposed to six months ago.

We should be proactive in our response and let him know we will be able to do something about this. Could you produce a draft letter to Lord A.

I hope things do not get nasty. Maybe we should review our responsibility for this and be up to speed about what we should do in case it does. Could you please let me have some notes regarding the issues we might face.

Many thanks

Michael Hart

Associate Director

ndg architects ltd

QUESTION 3
Page 1 of 1

Email@

Date: 25th February 2015

To: Candidate@ndgarchitects.com

From: Belinda@ndgarchitects.com

Re: Proposed Stables Conversion at Tally Ho

As you may know, we have just finished some major refurbishment work on the listed manor house at Tally Ho which also incorporated an assessment of the potential to install a bio-digester to provide renewable energy for the estate.

We seem to be doing something right as I have received an email from Lord Avafewbob about another possible project. He has read somewhere that you can convert existing farm buildings into residential units without the need for planning permission. He wants us to provide a quotation to provide building regulation drawings as he wants enabling works to the existing stable to start in a couple of weeks!!!!

I have some serious concerns about this as I feel his view cannot be correct.

So, as you are completing your Part 3 course I think this is a good opportunity for you to investigate his claims and analyse and highlight potential issues that need to be considered, namely, do we need planning permission? And, are there any consequences to his possible actions?

I would like you to write a letter explaining your thoughts on the proposal, its viability, potential next steps and immediate advice to the client.

Many thanks

Belinda Carter
Director

ndg architects ltd

QUESTION 4
Page 1 of 3

Email@

Date: 25th February 2015

To: Candidate@ndgarchitects.com

From: Belinda@ndgarchitects.com

Re: University of Madcaster

We have been asked by the University of Madcaster to submit a planning application for their new boat house on the river.

As you know, we produced some very quick sketches which resulted in a successful bid to Sport England.

The site is on the flood plain and is also within a conservation area and there is likely to be opposition from the neighbours who have gardens either side. A pre application submission will be required as well as a formal planning application.

You must assume that we will have to start from scratch with the design based upon their brief for a store for rowing eights at ground floor and changing rooms and a club room at first floor.

The value of the job is about £2m construction cost, but the University are not interested in us applying some sort of percentage fee. They want a break down of tasks with an estimate of professional time against each one in order to arrive at a lump sum fee.

They also require us to advise which other consultants we may need in order to submit a valid application and have asked if we can employ these companies as sub consultants.

In summary, could you please do the following

1. Draft a letter to the University Facilities Manager setting out our fee, basis of appointment, proposals for invoicing and our approach to the sub consultants.
2. Produce some sort of spread sheet (basically a list of key activities listed sequentially) identifying tasks and assuming all the work will be done by an architect at £60 per hour to justify the lump sum fee.
3. Add an amount for expenses (say £200) and VAT.
4. A list of the other consultants we wish to employ, but don't worry about their actual fees for now.

QUESTION 4
Page 2 of 3

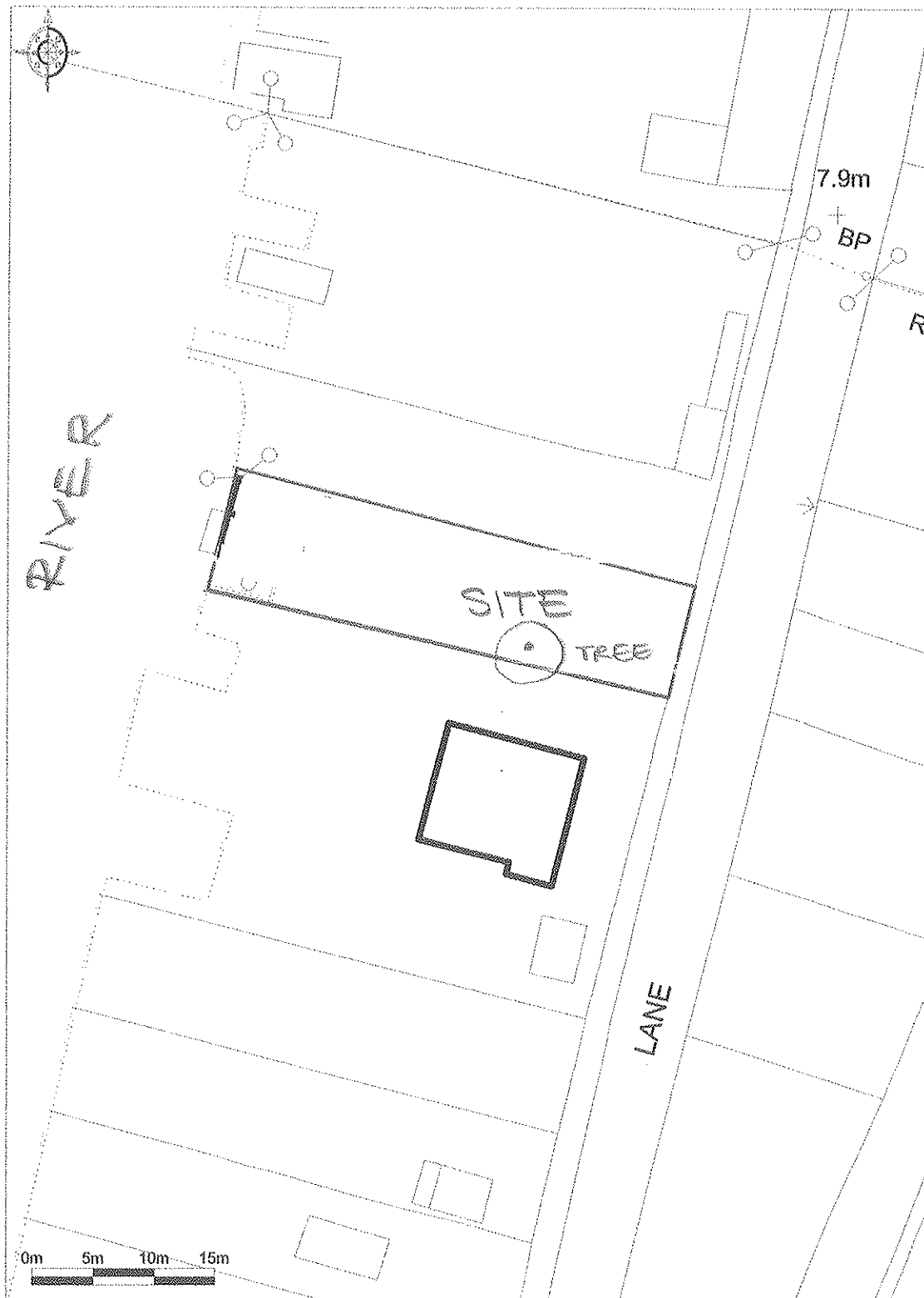
In addition I would be pleased to get your thoughts on the pros and cons of employing the other consultants as sub consultants and also any conditions you think we should highlight to the Facilities Manager at the University in terms of our lump sum fee.

Many thanks

Belinda Carter
Director

ndg architects ltd

QUESTION 4
Page 3 of 3



Email@

Date: 25th February 2015

To: Candidate@ndgarchitects.com

From: Sven@ndgarchitects.com

Re: Luxury Apartments in Madchester

You will remember last month's TV news and have seen the large cargo ship from the Far East which ran aground in the estuary during the bad weather. The ship is apparently carrying the three new boilers which were due on site.

The mechanical consultant has specified high performance equipment from a local company which boasts of the design and manufacturing skills of their engineers and we were all given the impression that the boilers were made locally. The supplier has now informed us that the boilers are on the listing ship and don't know when they will be recovered or if they are damaged.

We have also received a letter from the contractor, which is attached, informing us of the boiler problem which is likely to cause substantial delays and costs.

The JCT Standard Building Contract with Quantities 2011 has been used and the contractor is claiming that they have been misled by the mechanical consultant and are therefore refusing to accept any responsibility. They have however found alternative boilers which they can have delivered within 4 weeks. They are also requesting a formal architect's instruction confirming acceptance of the boiler alternatives along with a 9 week extension of time which will be the delay if the new boilers are accepted.

Can you please urgently forward me a memo with your thoughts on this as well as a draft reply to the attached contractor's letter.

Many thanks

Sven Berg
Architect

ndg architects ltd

Seedons

**Seedon House
Bridge Road
Madchester
M4 5FP**

**NDG Architects Ltd
Wharf Mill
Paccadilly
Madchester
MAD PPA**

23rd February 2015

Dear Sirs

Re: Luxury Apartments in Madchester

We wish to give you notice as required by building contract clause 2.27.1 that there is a substantial delay relating to the specified boilers and we cannot achieve the contract completion date. We are unable to suggest any revised completion date as required by clause 2.27.2 but this delay is a relevant event and we intend to submit a formal claim to extend the contract completion date and claim all costs which we incur as a result of this.

Your client's mechanical consultant has produced the entire design and specification for these boilers but has misled us to believe they were manufactured locally which turns out not to be the case. We have worked entirely in accordance with the consultants design and specification and our company will not accept any responsibility for the delay of equipment which has to be delivered from overseas without our knowledge. If we knew that equipment was being delivered from abroad we would have very strongly suggested that the specification was revised.

We have however been very proactive and wish to limit any inconvenience to your client. Our sub contractor has sourced alternative boilers which are capable of serving the building and can be available in 4 weeks and will limit the overall delay to the completion date to 9 weeks. If you would like us to proceed with this then we will require a formal Architect's Instruction to amend the contract specification, extend the completion date by 9 weeks and increase the contract sum by £105,000.

We await your response.

Yours faithfully
Dave Evans
Contractor

END OF DAY ONE

**Please ensure that you hand a copy of your draft answers
to your exam supervisor by 5.30pm**



RIBA PART 3
EXAMINATION IN PROFESSIONAL PRACTICE

**February 2015
Practice Problems**

Day 2

Questions 6-10

QUESTION 6
Page 1 of 1

Email@

Date: 26th February 2015

To: Candidate@ndgarchitects.com

From: Sue@ndgarchitects.com

Re: Office management - BIM

As you know the office has invested a considerable amount of money in Revit software. We have to make a commitment to BIM because of all the public funded work that we undertake. Most of the D and B contractors involved in the schools programme are also insisting on this.

We will be discussing CAD generally at the next Directors meeting and I would be grateful if you could put together a brief presentation to senior managers. As you know, some of our older architects are very sceptical and, although understand CAD, do not realise how important BIM is and how it will affect the work of the office

What I need is a document that sets out the key issues, how BIM works in the broadest sense, how it is used by Architects, Engineers and other consultants. In addition explain why contractors like it so much and how the subcontractors fit in. Finally, suggest briefly how we can sell ourselves to Clients as being BIM proficient in terms of the benefits to them

Please note, I want you to direct your thoughts specifically to our Directors and what is important from their perspective. I do NOT want a cut and paste from the internet (I could do that myself!). Feel free to use a bullet point format with key headings

Many thanks

Sue Allen
Associate Director

ndg architects ltd

QUESTION 7
Page 1 of 2

Email@

Date: 26th February 2015

To: Candidate@ndgarchitects.com

From: Belinda@ndgarchitects.com

Re: Amelia's Bakery

You saw the headline in the local press some time ago when the frontage of the Amelia's Bakery came down injuring two pedestrians.

Well we are being drawn into the litigation that has followed this event.

James is very worried indeed and we need a clear head to examine whether we are liable or not, before we sit down with our P.I. advisers, their Loss Adjusters and Lawyers.

You will be aware of the discussions in the office but let me explain what has happened and the history of work at these premises.

The claimants are the two pedestrians who were walking past the shop when the concrete façade fell off and seriously injured them. They have sued Amelia's Bakery, the company that fitted out the shop and ourselves.

Let me provide the facts.

- Three years ago the shop front was remodelled by shop designers, Facades & Co. Work was carried out to the fascia supports to allow for a retractable awning housing in a wooden box to be installed above the shop windows.
- Two years ago the landlord, ABC Properties arranged for building work to be carried out to the flats above the shop. We acted for them as architects, which included contract administration.

QUESTION 7
Page 2 of 2

- Last year Amelia Baker, the tenant, experienced problems in using the awning. She believed the awning box had been knocked by ABC Properties workmen during the building work.
- We then made a site visit at ABC Properties request, to investigate Amelia's complaint, and reported that the problem appeared to be with the awning's retracting mechanism.
- In January this year, the shop front detached causing serious injuries to two passers-by.

Amelia and Facades & Co allege that we either failed to see, or failed to appreciate from what we saw that the entire shop façade was in a dangerous condition, with the result that we failed to advise that urgent action was required.

We need to assess these issues objectively in order to present a robust defence. I'm looking at this myself but could you do the same, so that we can have an early meeting (with James) to establish our response to the claim against us.

I look forward to receiving a report from you on this matter which explores our liability or not. This should also propose what you believe to be our course of action.

We can then compare notes in preparation for our meeting with the loss adjuster and our insurer's lawyers.

Many thanks

Belinda Carter
Director

ndg architects ltd

QUESTION 8
Page 1 of 1

Email@

Date: 26th February 2015

To: Candidate@ndgarchitects.com

From: James@ndgarchitects.com

Re: CDM regulations

As you know the CDM regs have changed and there is a new term “principal designer” which seems to mean that we, as architects, will be coordinating everything from a health and safety point of view. There are also changes for domestic projects.

You are probably the best informed member of the office as you revise for your Part 3 exam so can you please unpick the advice on the HSE web site and explain what is going on!

1. Specifically, can we just sub contract our new “Principal Designer” role to “Safe and Sound CDM Services”, the firm we have always used for CDM matters? If so, are there any problems or risks associated with this arrangement and how would appointments be dealt with pre and post construction?
2. With regard to the jobs we do for domestic clients including one off houses, do we need to amend our appointment document to reflect the new regulations and will this have a bearing on the fee we need to charge?
3. Finally, with regard to the school job you are working on, we already have “Safe and Sound CDM Services” appointed. This project has just gone out to tender but will certainly not be finished on site by 6th April, what will we need to do to comply with the new arrangements?

Many thanks

James Noble
Consultant

ndg architects ltd

QUESTION 9
Page 1 of 1

Email@

Date: 26th February 2015

To: Candidate@ndgarchitects.com

From: Belinda@ndgarchitects.com

Re: Marketing strategy

As you know we are extremely busy at present and have a good programme of work for the future. However I am told this is the very time a Marketing strategy and action plan should be developed.

Being new to the practice you will have a fresh and critical eye about how our marketing is currently carried out in comparison with other practices you have worked in or those of your fellow graduates. You will also have learnt about different aspects, methods and delivery options during your Part 3 course; this is an ideal opportunity for you to apply theory to live practice.

I have to make a presentation to the other directors on this topic and would be grateful if you could also come along and be ready to answer questions

In order to get the ball rolling, could you prepare a draft for a Marketing Strategy for the practice; just bullet point format is fine, to form the basis for a policy document. It should include some simple definitions; there are so many jargon words bandied around that many are misunderstood: marketing; promotion; PR/ public relations; client relationships; online marketing and communications. It should refer to overall aims: what needs to be done, and over what period of time. How will we know at the end of one year / two years that it has been effective? What staff resources will be required to carry out this exercise?

Many thanks

Belinda Carter

Director

ndg architects ltd

QUESTION 10
Page 1 of 1

Email@

Date: 26th February 2015

To: Candidate@ndgarchitects.com

From: Sue@ndgarchitects.com

Re: Madchester Academy – proposed Performing Arts Block

Great news! Last night the Governors of Madchester Academy approved our designs for the new Performing Arts Block and the works budget of £3m.

The Academy has secured funding from the Government's Education Funding Agency and has requested our advice on how to identify potential tenderers for the project. As this project is receiving public funding it is essential that the procurement process is fair and transparent.

Can you prepare a short note of the options available when drafting a list of potential tenderers for this project and the pro's and con's of the various options.

Can you also identify the key factors and checks we should carry out relating to the potential contractors before recommending their suitability for the tender list.

The Principal also told me that one of their Governors owns a construction company and he has asked if his firm can be included on the tender list. Can you jot down some issues which I can discuss with the Principal, bearing in mind the Board of Governors is our client for this project.

Many thanks

Sue Allen
Associate Director

ndg architects ltd

END OF DAY TWO

**Please ensure that you hand a copy of your draft answers
to your exam supervisor by 5.30pm**

DAY THREE

CHECK LIST – DO NOT FORGET TO SEND

DUPLICATE COPIES OF

C.V.

SELF EVALUATION

PEDR SHEETS

CASE STUDY

AND RESPONSES TO PRACTICE PROBLEMS

CD OF SUBMISSION (this can be sent by 2nd March 2015 if necessary)

TOGETHER WITH

**DECLARATION – signed by yourself and the Exam Supervisor
Sealed draft copies of each day's responses**

**REMEMBER to obtain a Certificate of Posting and keep in a safe place to
present at interview.**